Chapter 04 Homework Answer Sheet

Name:__________Student ID:__________Instructor:__________Score:___________

Part A multiple choice (11 points)

1-5： BDCCD

6-10： BBCD D/C

11： B

Part B short question (24 points)

12 (5 points)

(**a**) $2CH_3COOH(aq) + Ba(OH)_2(aq) \rightarrow Ba(CH_3COO)_2(aq) + 2H_2O(l)$

$CH_3COOH(aq) + OH^-(aq) \rightarrow CH_3COO^-(aq) + H_2O(l)$

(**b**) $Cr(OH)_3(s) + 3HNO_2(aq) \rightarrow Cr(NO_2)_3(aq) + 3H_2O(l)$

$Cr(OH)_3(s) + 3HNO_2(aq) \rightarrow Cr^{3+}(aq) + 3NO_2^-(aq) + 3H_2O(l)$

(**c**) $HNO_3(aq) + NH_3(aq) \rightarrow NH_4NO_3(aq)$

$H^+(aq) + NH_3(aq) \rightarrow NH_4^+(aq)$

(**d**) $CdS(s) + H_2SO_4(aq) \rightarrow CdSO_4(aq) + H_2S(g)$

$CdS(s) + 2H^+(aq) \rightarrow Cd^{2+}(aq) + H_2S(g)$

(**e**) $MgCO_3(s) + 2HClO_4(aq) \rightarrow Mg(ClO_4)_2(aq) + H_2O(l) + CO_2(g)$

$MgCO_3(s) + 2H^+(aq) \rightarrow Mg^{2+}(aq) + + H_2O(l) + CO_2(g)$

13 (4 points)

(a) +6, (b) +5, (c) +5, (d) +2

14 (4 points)

(**a**) Exchange (Metathesis) Reactions

(**b**) $K_3PO_4(aq)+3AgNO_3(aq)\rightarrow Ag_3PO_4(s)+3KNO_3(aq)$

(**c**) To determine the limiting reactant, we must examine the number

$n(K_3PO_4)= 70.5\times10^{-3}g\div212.2695g/mol=3.32\times10^{-4}$ mol

$n(AgNO_3)=15.0\times10^{-3}L\times0.0500mol/L=7.50\times10^{-4}$ mol

Comparing the amounts of the two reactants, we find that there

1 mol K_3PO_4 ≌ 2.26 mol $AgNO_3$

According to the part (b), 1 mol K_3PO_4 ≌ 3 mol $AgNO_3$.

Thus, there is insufficient $AgNO_3$ to consume the K_3PO_4.

Therefore, $AgNO_3$ is limiting reactant

(d)

percentage yield $=84.2\div(15.0\times10^{-3}\times0.0500\div3\times418.5792\times1000)\times100\%=80.5\%$

15 (2 points)

$PbS(s)$ and $PbCl_2(s)$ will precipitate.

16 (4 points)

(**a**) acid, mixture of molecules and ions, weak electrolyte

(**b**) none of the above, molecules , nonelectrolyte

(**c**) salt, ions, strong electrolyte

(**d**) base, ions, strong electrolye

17 (5 points)

(**a**) $Ni(s) + Cu(NO_3)_2(aq) \rightarrow Cu(s) + Ni(NO_3)_2(aq)$

(**b**) NR

(**c**) NR

(**d**) $2Cr(s) + 3CoCl_2(aq) \rightarrow 3Co(s) + 2CrCl_3(aq)$

(e) $H_2(g) + 2AgNO_3(aq) \rightarrow 2Ag(s) + 2HNO_3(aq)$